



Xerox Phaser 3116

If you need to print a large volume of documents, but do not really need photo-quality printouts, the Canons are perhaps your best bet—they'll save you a great deal of time. At an average rate of 3 seconds saved per print, your employees will

wait 3,000 seconds less for every 1,000 documents printed—or more succinctly, close to an hour of work time saved per 1,000 prints!

Since these printers are meant for offices, they will be called on to print documents containing graphs, charts and text. We also tested all the printers with our combi-document.

Here, the Xerox 3116, with a score of 12.43 seconds, narrowly beat the Canon LBP2900 and LBP 3200 with scores of 12.55 and 12.57 seconds respectively. Coming in last, again, was the WeP 1600 with a time of a little over 16 seconds. Quantifying this further, a 4-second deficit means 66 minutes wasted for every 1,000 prints!

Surprisingly, apart from the WeP and Xerox, all printers were even faster when printing in 'Best' quality mode, returning faster scores than when printing in Normal mode!

We really were disappointed with the WeP 1600, which took over 27 seconds to churn out the page. This is probably due to a slower buffer memory, but the fact remains that it took twice as long as the Canon LBP 3200.

Speed, however, is not always the deciding factor when buying a printer. No client will be

How We Tested

Our test bed comprised a Pentium 4 3.2 GHz processor plugged onto an MSI 75P Neo FISR2 motherboard with 512 MB of Corsair 400 MHz DDR RAM, a Gainward nVidia GeForce FX 5950 graphics card, and a 250 GB 7200 rpm SATA Maxtor hard disk. We used Windows XP with SP1 as the OS, and the system was loaded with the latest chipset and graphics drivers. We also installed the USB patch for XP and connected the devices on the USB 2.0 port for optimal performance. We used 100 g/sm paper from Berga (A4 size) for printing.

Categorisation

The printers were categorized as colour printers and monochrome printers.

We further divided monochrome printers into two sub-categories: those with duty cycles of less than 15,000 (SoHo) and those with more than 15,000 duty cycles (Workgroup).

Laser printers

Features: Printers were rated on various features: Duty cycle, memory card readers, duplex printing, input and output tray capacity, input buffer, maximum resolution supported by the printers, and so on. They were rated on the basis of features they supported.

Performance: We tested the performance on three parameters; speed, quality, and efficiency.

Speed and Quality Test: We used a text document to test the raw print speeds of the printer. The quality settings used were Normal and Best. To test their ability to handle various aspects of a regular document, we created a combi-document with black text, interspersed with graphs and images. We clocked the time it took them to print this document in normal and best quality modes, and analysed the print quality. The final test for colour printers was the photo test, where we printed an A4 size image. We clocked the time taken to complete the print and examined the quality of the print. The settings used here were the highest quality available.

Laser MFDs: MFDs have a scanner, printer and copier. Some even have fax facility. MFDs with fax facility earned bonus points.

Features: The printer part of the features that were noted remained the same as in the laser printers. In the scanner part, the maximum scan resolution, bit depth, scanner type, scanning element, scanning area, etc., were noted. In the copier and fax

part, the ability to work as a standalone device, maximum number of copies in multi-copy mode, memory for storing fax pages, and more, were noted.

Printer Test: The test process here was exactly the same as that of the laser printers.

Scanner Test: We tested the scanners for speed using an average of five previews of an A4 size image. This was done when the scanner was fresh out of the box to incorporate the warm-up time. Scanners with a higher warm-up time didn't fare as well as the others. We also tested the warmed up scanners to negate any possibility that a scanner might scan faster than the others once warmed up.

We imported the same A4 size image at 150 dpi and 600 dpi, in Adobe Photoshop, and scanned a full text document in B&W mode at 200 dpi. The scanner then underwent a test to determine its colour differentiating abilities using a Kodak IT8 card with different shades of colour—each with a little difference from the preceding one. Next, we tested the resolution card to figure actual scanning capabilities by checking for its native resolutions. We scanned the card at 300 dpi at threshold colour depth to see how many boxes it could differentiate in between the lines. Finally, we did an OCR test to check the scanner's ability to differentiate light and dark areas. A print article was scanned into ABBYY FineReader Pro 7.0, and verified for wrongly-interpreted letters in the software. We then gave a percentile value to the devices.

Copier

We copied a text document in B&W and rated the MFDs on their speed and quality of the printout.

Warranty and Price: The number of years of warranty, type of warranty, number of authorized service centres, price of the printer and the toner and the rated cost per page were also taken into account and rated accordingly.

How the Awards Are Given

The scores from features, performance and price were given weightages suitable to the category. An overall score out of 100 is calculated. The product that scores the highest here is adjudged the winner of the Best Buy Gold award for this category. The second highest score gets the Digit Best Buy Silver award.